SwiftLog Executive Summary

Machine Learning Track | Week 5

Objective

Build an educational machine-learning prototype to explore delivery delays, product returns, weekly product demand, and next-month customer revenue using the supplied SwiftLog datasets.

What was completed

Reviewed source tables; engineered task-specific features; compared candidate algorithms; evaluated held-out predictions; saved selected model pipelines; generated comparison tables and evaluation visuals; and prepared a Streamlit demonstration app.

Results at a glance

Problem	Algorithm
Delivery Delay Classification	Random Forest (tuned)
Product Return Classification	Decision Tree
Product Demand Forecasting	Random Forest Regressor (tuned)
Customer Value Prediction (Next Month)	Linear Regression

What the results mean

The current models are not ready for operational use. The delay classifier has high F1 but ROC-AUC close to 0.5, suggesting weak class separation. The return classifier also shows weak discrimination. Demand and next-month customer-value regression have negative held-out R-squared, so they do not outperform a simple mean baseline on the test period.

Recommended next steps

Collect richer historical data, verify target definitions, add operationally available predictors, compare against simple baselines, and validate on later time periods. Use the app only as a demonstration and keep human review in the decision process.

Conclusion

The project delivers a documented prototype workflow and evaluation. The most important outcome is identifying the data and validation improvements needed before any business deployment.